

LOOMX

WLED UI Replacement Specification

Custom web UI · Parent/Child architecture · WebSocket sync

Mission

Replace the entire stock WLED web UI with a custom, brand-aligned, operator-grade interface for the Loomx Lighting platform. This document is the complete specification — pages, components, copy, behavior, brand language, and acceptance criteria. Claude Code should treat this as the authoritative source of truth for the UI layer.

What this is NOT: A patch on top of WLED's existing UI. The stock `/wled00/data/` assets are deleted and replaced wholesale. The C++ HTTP and WebSocket endpoints stay (with additions); only the front-end is swapped.

Project Context

Loomx Lighting is a custom LED design and installation company based in Melville, NY. This firmware powers a self-contained Parent/Child controller stack for synchronized LED installations in nightclubs, venues, and hospitality environments. The Parent controller hosts its own WiFi access point; Children connect to it over WebSocket; operators control everything from a phone, tablet, or laptop through the web UI served by the Parent.

Audience

Primary user	Loomx operator running the system at a venue (DJ booth, FOH, install crew)
Device	Phone or tablet (iPad most common). Desktop browser also supported.
Expertise	Mid-level. Knows what an LED is, doesn't want to think about GPIO pins.
Session	Often standing, in low light, sometimes wearing gloves. Touch targets must be generous.

Brand Identity

The UI is dark-mode-only. Pure black backgrounds, a single neon-green accent, and restrained typography. No gradients. No drop shadows. No decorative chrome. The interface should feel like professional broadcast or DJ gear — not like consumer home automation.

Color Tokens

Background	#000000 — Pure black. Used for app background.
Surface	#0A0A0A — Cards, modals, sheets.
Surface elevated	#1A1A1A — Inputs, buttons (default state).
Border default	#2A2A2A — Subtle dividers, card borders.
Border active	#C6F542 — Selected/focused elements.
Text primary	#FFFFFF — Headings, primary copy.
Text secondary	#888888 — Body text, hints, labels.
Text tertiary	#444444 — Disabled, micro-labels, footers.
Accent (brand)	#C6F542 — Neon green. Primary actions, active states, brand mark, live indicators.
Accent ink	#000000 — Text on accent backgrounds (primary buttons).
Success	#5DCAA5 — Online status dots, connected indicators.
Danger	#F09595 — Destructive action labels, error states.
Warning amber	#EF9F27 — Pending state, reconnecting.

Typography

Font family	'Inter', system-ui, -apple-system, sans-serif
Display / Brand	'Inter', font-weight: 900. Used only for the LOOMX wordmark on splash and headers.
Heading L	22px / 28px line / weight 500 / white
Heading M	18px / 24px line / weight 500 / white

Heading S	15px / 20px line / weight 500 / white
Body	14px / 20px line / weight 400 / #888 (secondary)
Body strong	14px / 20px line / weight 500 / white
Label	12px / 16px line / weight 500 / #888 — uppercase, letter-spacing 0.5px
Micro	11px / 14px line / weight 400 / #444 — bottom-of-screen meta only
Mono / IDs	'JetBrains Mono', ui-monospace, monospace — for MAC addresses, IDs

Spacing & Radius

Outer page padding	24px on phones, 32px on tablets
Card radius	16px
Button radius	14px
Input radius	12px
Pill / chip radius	999px (fully rounded)
Touch target min	44px height — non-negotiable
Spacing scale	4 · 8 · 12 · 16 · 20 · 24 · 32 · 48 px

Iconography

Use Lucide (lucide-react or lucide as web component). Outline only, never filled. Default size 20px inline, 24px for primary actions, 18px for inline-with-text. Icon color matches the text it sits next to — accent green only when the action itself is accent.

Motion

- Transitions: 150ms ease-out for hover/focus, 200ms for view changes.
- Splash loader: 3-dot sequential pulse, 1.4s loop.
- Active sequence button: 2px accent border pulse, 1s loop.
- Respect prefers-reduced-motion — disable all loops, keep transitions.
- Never use bounce, spring, or elastic easings. Strictly linear or ease-out.

Front-End Architecture

Stack

Framework	Svelte (preferred) or Preact. Small bundle size is critical — must fit in ESP32 flash.
Build tool	Vite. Output: single bundled, gzipped HTML+CSS+JS payload.
Styling	Vanilla CSS with CSS custom properties for tokens. No Tailwind, no CSS-in-JS runtime.
Icon library	Lucide (tree-shaken — only import icons used).
State	Svelte stores (or Preact signals). No Redux, no Zustand.
Routing	Hash-based client router (svelte-spa-router or custom 30-line router).
Data layer	Single WebSocket connection to Parent + REST for non-realtime CRUD.
Bundle target	< 80 KB gzipped total (HTML + CSS + JS).

Build Output

Vite produces a single index.html with inlined CSS and JS, gzipped. A post-build script converts the gzipped bundle into a C header file (web_ui_bundle.h) that gets compiled into firmware. The Parent's HTTP handler serves it with Content-Encoding: gzip.

```
// Firmware HTTP route (pseudo-code)
server.on("/", HTTP_GET, [](AsyncWebServerRequest *req) {
    AsyncWebServerResponse *resp = req->beginResponse_P(
        200, "text/html", web_ui_bundle, web_ui_bundle_len
    );
    resp->addHeader("Content-Encoding", "gzip");
    resp->addHeader("Cache-Control", "public, max-age=3600");
    req->send(resp);
});
```

Directory Layout

```
web-ui/
+-- public/           # Static assets (favicon, etc.)
+-- src/
|   +-- main.js       # Entry — mounts root component
|   +-- App.svelte    # Root, handles routing
|   +-- routes/       # One file per page
|       +-- Splash.svelte
|       +-- RoleSelect.svelte
|       +-- ParentSetup.svelte
|       +-- ChildStandby.svelte
```

```

| | +-- ViewMode.svelte
| | +-- EditMode.svelte
| | +-- tabs/
| |   +-- Sequences.svelte
| |   +-- SequenceBuilder.svelte
| |   +-- Presets.svelte
| |   +-- Children.svelte
| |   +-- Configs.svelte
| |   +-- Settings.svelte
| +-- components/          # Reusable UI primitives
| | +-- Button.svelte
| | +-- Card.svelte
| | +-- Toggle.svelte
| | +-- Input.svelte
| | +-- Select.svelte
| | +-- ColorPicker.svelte
| | +-- Toast.svelte
| | +-- Modal.svelte
| | +-- BrandMark.svelte   # LOOMX wordmark
| | +-- Loader.svelte      # 3-dot pulse
| | +-- StatusDot.svelte
| | +-- SignalBars.svelte
| +-- stores/              # Svelte stores
| | +-- role.js            # 'parent' | 'child' | null
| | +-- network.js         # SSID, password, hidden, channel
| | +-- children.js        # List of bound children
| | +-- presets.js         # Preset CRUD
| | +-- sequences.js       # Sequence CRUD
| | +-- configs.js         # Per-child LED configs
| | +-- ws.js              # WebSocket connection state
| | +-- toasts.js          # Notification queue
| +-- lib/
| | +-- api.js             # REST client
| | +-- ws.js              # WebSocket client
| | +-- format.js          # MAC, time, duration formatters
| +-- styles/
| | +-- tokens.css         # All CSS custom properties
| | +-- global.css         # Resets, base styles
+-- index.html
+-- vite.config.js
+-- package.json
+-- scripts/
    +-- bundle-to-header.js # Converts dist/ to C header

```

Page Specifications

Each page is documented with: purpose, route, all copy verbatim, ASCII layout, components used, state read/written, and acceptance criteria. Build pages in this order.

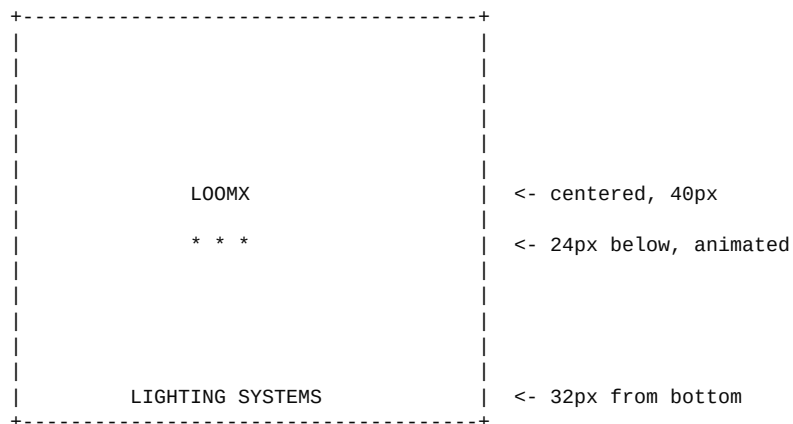
Page 0 — Splash / Startup

Route	/ (initial mount, before role resolution)
Component	routes/Splash.svelte
Purpose	Brand moment on every boot. Holds while firmware initializes and role is resolved.
Min duration	1500ms (even if state resolves faster — branding is intentional)
Next route	/setup/role if no role set · /app if Parent · /child if Child

Copy

- Logo: **LOOMX** (neon green wordmark, centered, ~40px tall)
- Loader: three dots, neon green, sequential pulse animation
- Footer: **LIGHTING SYSTEMS** (11px, uppercase, letter-spacing 2px, color #444)

Layout



Acceptance

- Screen is pure black (#000) from edge to edge — no surface card.
- Logo perfectly centered both axes.
- Loader dots pulse: 100%, 60%, 30% opacity rotating with 200ms stagger.
- Transition out: 200ms fade to next route.
- Hides browser scrollbars during display.

Page 1 — Role Selection (First Boot Only)

Route	/setup/role
Component	routes/RoleSelect.svelte
Purpose	User picks Parent or Child. Stored to NVS, drives all subsequent UI.
Next route	Parent -> /setup/network · Child -> /child
Re-entry	Reachable later via Settings -> Role

Copy (verbatim)

- Header brand: **LOOMX** (small, top-center)
- Title: **Set up this controller**
- Subtitle: *Pick a role to get started. You can change this later.*
- Parent card title: **Parent**
- Parent card body: *Hosts the network. Controls children. No LEDs.*
- Parent card icon: broadcast (Lucide)
- Child card title: **Child**
- Child card body: *Drives LEDs. Receives commands from a parent.*
- Child card icon: lightbulb (Lucide)
- Primary button: **Continue**
- Footer: *Role: not set* (until selection made), then *Role: Parent / Role: Child*

Layout

LOOMX Set up this controller Pick a role to get started. You can change this later.	<- 22px logo, top <- 22px white <- 14px gray
[[P]] Parent -> Hosts the network. Controls children. No LEDs.	<- Selected: 1px green border Icon bg = green 15% Title in green
[[C]] Child -> Drives LEDs. Receives commands from a parent.	<- Default: 0.5px #2A border
Continue	<- Accent green bg black text
Role: not set	

Interaction

- Tapping a card sets selection. Only one selected at a time.
- **Continue** button is disabled (opacity 40%) until a selection is made.
- On Continue, store role to NVS, then route forward.
- If user backs out before Continue, no state is written.

Page 2 — Parent Setup Wizard

Route	/setup/network
Component	routes/ParentSetup.svelte
Purpose	Confirm or customize auto-generated network credentials.
Next route	/app

Auto-generation Rules

- SSID format: Loomx_Parent_NNNN where NNNN is a random 4-digit number.
- Password format: Loomx + 4-digit number + 1 color word from a fixed list. Example: Loomx2847Blue.
- Color word list: Blue, Red, Green, Pink, Teal, Coral, Amber, Lime, Cyan, Violet.
- Hidden network: off by default.
- Channel: Auto by default (firmware picks least-crowded 2.4 GHz channel on boot).

Copy

- Title: **Your network is ready**
- Subtitle: *Auto-generated. Edit now or later.*
- Field label: **Network name**
- Field label: **Password**
- Toggle label: **Hide network**
- Toggle hint: *Network won't appear in scans. You'll need to enter the name manually to connect.*
- Regen button: refresh-cw icon, no text
- Primary button: **Continue**
- Secondary link: *Skip — use defaults*

Layout

```

+-----+
|                LOOMX                |
| Your network                          |
| is ready                            |
| Auto-generated. Edit now or later.   |
|                                     |
| NETWORK NAME                        |
| +-----+ +-----+                 |
| | Loomx_Parent_4827 | | @ |         |
| +-----+ +-----+                 |
|                                     |
| PASSWORD                          |
| +-----+ +-----+                 |
| | Loomx2847Blue    | | @ |         |
| +-----+ +-----+                 |
|                                     |
| Hide network [ OFF ]                |
| Network won't appear in scans.      |
|                                     |
| +-----+

```

		Continue		
	+	-----	+	
		Skip – use defaults		
+		-----	+	

Page 3 — Main App, View Mode (Default Operator Screen)

Route	/app
Component	routes/ViewMode.svelte
Purpose	The operator's primary interface. Minimal. One-tap sequence and preset triggering.
State read	sequences, presets, children (count + online status), running sequence
Default	This is the landing page for Parent every boot after first setup.

Top Bar

- Left: **LOOMX** wordmark (16px, neon green)
- Center: ** 4 nodes online* — green dot if all online, amber if any reconnecting, red if any offline. Tappable -> opens Children tab in Edit Mode.
- Right: gear icon (settings) -> enters Edit Mode

Section Headers

- **SEQUENCES** (12px, uppercase, letter-spacing 0.5px, #888)
- **PRESETS** (same style)

Sequence Buttons

- 3-column grid, 8px gap.
- Each button: 56px tall, surface bg (#2A2A2A), 0.5px border (#3A3A3A), 13px white label, radius 12px.
- Active (currently running): 1.5px accent border, label in accent green, subtle pulse animation.
- Long press (500ms): opens action sheet — *Edit · Duplicate · Export · Delete*

Preset Buttons

- Same grid and size as sequence buttons.
- Optional: small color swatch row inside the button (8px squares, one per child) for quick visual ID.
- Long press: *Edit · Rename · Delete*

Stop All Button

- Pinned at bottom, full width, 44px tall.
- Transparent background, 0.5px danger-tinted border (#5A3030), text in #F09595.
- Action: sends stop_all WebSocket broadcast to all children, returns lights to off.
- Confirmation: only required if a sequence is actively running, otherwise instant.

Layout

```

+-----+
| LOOMX   * 4 nodes online  [[*]] |
| ----- |
|                                     |

```

SEQUENCES		
+-----+	+-----+	+-----+
Entrance	Countdown	Drop
+-----+	+-----+	+-----+
+-----+	+-----+	
Sunset	Strobe	
+-----+	+-----+	
PRESETS		
+-----+	+-----+	+-----+
# All Rd	# All Bl	Mix 1
+-----+	+-----+	+-----+
+-----+		
Stop all		
+-----+		
+-----+		

Empty States

- No sequences: *No sequences yet. Tap [[*]] to create one.*
- No presets: *No presets saved. Tap [[*]] to capture the current state.*
- No children online: *No children online. Tap to manage.* (top bar pill becomes amber)

Page 4 — Edit Mode (Tabbed)

Route	/edit/:tab — tabs: sequences presets children configs settings
Component	routes/EditMode.svelte + tab components
Purpose	All authoring, configuration, and management. Hidden behind the gear icon.
Default tab	sequences

Persistent Chrome

- Top: back arrow (returns to View Mode) + label **Edit mode** + small connection indicator.
- Below: horizontal scrollable tab bar — Sequences · Presets · Children · Configs · Settings.
- Active tab: white text, accent green underline (2px) below text.
- Inactive tabs: #888 text, no underline.

Layout

```
+-----+
| [<-] Edit mode * 4/4 |
| ----- |
| Sequences Presets Children Configs ... |
| ----- |
|                                     |
| (tab content here) |
| ----- |
+-----+
```

<- active tab underline

Page 4a — Edit Mode › Sequences

Route	/edit/sequences
Component	routes/tabs/Sequences.svelte
Purpose	List, create, edit, delete, import, export sequences.

Copy

- Page title: **Sequences**
- Top-right actions: **+ New** button, **Import** button (file picker for .json)
- Row format: *{name} {step count} steps {total duration} [... menu]*
- Row menu: *Edit · Duplicate · Export · Delete*
- Empty state: *No sequences yet. Tap + New to build your first one.*

Layout

```
+-----+
| Sequences [+ New] [Import] |
| +-----+ |
| | Entrance 5 steps 12s [...] | |
| | Countdown 3 steps 10s [...] | |
| | Drop 8 steps 30s [...] | |
+-----+
```

| +-----+ |
+-----+

Page 4a.1 — Sequence Builder

Route	/edit/sequences/:id (or /new)
Component	routes/tabs/SequenceBuilder.svelte
Purpose	Visually compose a sequence step by step.

Header

- Name input: *Sequence name* placeholder, 16px white.
- Loop toggle: **Loop** label + toggle.
- Right side: **Save** (primary accent) and **Cancel** (text button).

Step Card

Each step is a card containing:

- Step number badge (top-left): Step 1 in small accent pill.
- Target selector: *All children · Per child · Group A · Group B*
- If *Per child* selected: child color picker row below — one swatch per bound child.
- Effect selector: *Solid · Flash · Fade · Pulse · Chase · Rainbow · Custom*
- Color picker (when target is All or a group, single color).
- Duration slider: *0.1s — 60s*, also accepts typed input.
- Fade/transition input: *0.0s — 5.0s*
- Drag handle (left edge) for reordering.
- Delete button (right edge): red trash icon.

Footer

- **+ Add step** button — full width, dashed border, accent text.
- Total duration readout: *Total: 12.4s · 5 steps*

Layout

```
+-----+
| Name: [ Countdown ] |
| Loop: [ OFF ] [ Save ] [ X ] |
| ----- |
| |
| += Step 1 ===== x + |
| | Target: [ All children v ] | |
| | Effect: [ Solid v ] | |
| | Color: [ # white ] | |
| | Duration: [ ---*----- ] 3.0s | |
| | Fade: [ *----- ] 0.2s | |
| +=====+ |
| |
| += Step 2 ===== x + |
| | Target: [ All children v ] | |
| | Effect: [ Flash v ] | |
| | Color: [ # red ] | |
| | Duration: [ --*----- ] 2.0s | |
| +=====+ |
| |
```



```
| + + Add step -----+ |
| +-----+
|
| Total: 5.2s · 2 steps
+-----+
```

Page 4b — Edit Mode › Presets

Route	/edit/presets
Component	routes/tabs/Presets.svelte
Purpose	Manage the preset library.

Copy

- Title: **Presets**
- Actions: **+ Capture** (snapshot current live state) · **Import**
- Row: preset name, color swatch strip (one per bound child), [...] menu
- Row menu: *Apply* · *Rename* · *Export* · *Delete*
- Empty state: *No presets yet. Set up the state you want, then tap + Capture.*

Layout

```
+-----+
| Presets      [+ Capture] [Import] |
|
| +-----+
| | All Red    #####          [...] |
| | Mix 1      ##.##          [...] |
| | Sunset     #####          [...] |
| | All Blue   #####          [...] |
| +-----+
+-----+
```

Page 4c — Edit Mode › Children

Route	/edit/children
Component	routes/tabs/Children.svelte
Purpose	Discover, bind, unbind, and view status of children.

Sections

- **Bound (N)** — currently bound children, with live status.
- **Available to bind** — appears after Scan, lists nearby LumixLeaf APs.

Bound Row

- * green dot if online, o gray if offline.
- Name (editable on long press).
- MAC address in mono font, #888.
- Signal bars (|||| to).
- [...] menu: *Configure* · *Push config* · *Rename* · *Identify* · *Unbind*
- **Identify** action: flashes all LEDs on that child white for 3 seconds — for physical identification.

Discovery Row

- Checkbox on the left.
- SSID (mono font).
- Signal bars.
- Bottom bar appears when ≥ 1 selected: **Bind selected (N)**.

Layout

```
+-----+
| Children           [Scan for nodes] |
|                                     |
| BOUND (4)                                     |
| +-----+                                     |
| | * Child 1  AA:BB:CC:01 |||| [...] | |
| | * Child 2  AA:BB:CC:02 |||. [...] | |
| | * Child 3  AA:BB:CC:03 |||| [...] | |
| | o Child 4  AA:BB:CC:04 ---- [...] | |
| +-----+                                     |
|                                     |
| AVAILABLE TO BIND                                     |
| +-----+                                     |
| | ■ LumixLeaf_8821    |||. | |
| | ■ LumixLeaf_3047    ||.. | |
| +-----+                                     |
| +-----+                                     |
| |           Bind selected (0)           | |
| +-----+                                     |
+-----+
```

Page 4d — Edit Mode › Configs

Route	/edit/configs
Component	routes/tabs/Configs.svelte
Purpose	Configure each child's LED outputs. GPIO numbers never shown.

Critical Rule

Never display GPIO numbers in the UI. Outputs are always labeled *Output 1*, *Output 2*, etc. The GPIO mapping lives in firmware, keyed by board type, and is invisible to the operator.

Layout

- Child selector dropdown at top.
- List of outputs (count varies by child hardware — 4 or 8).
- Per output: LED type select, LED count input, color order select.
- Defaults: WS2812B, 60 LEDs, GRB.
- Footer actions: **Save to child** (accent), **Clone to...**, **Export**, **Import**

Clone Flow

- Tap **Clone to...** -> modal with checklist of all other bound children.
- Select one or more.
- Confirm — applies current child's config to all selected targets.
- Toast: *Cloned config to 3 children*

Layout

```
+-----+
| Configs                                     |
| Child: [ Child 1 v ] [Export][Import]    |
|                                           |
| +- Output 1 -----+                     |
| | Type: [ WS2812B v ]                     |
| | Count: [ 60 ]                           |
| | Order: [ GRB v ]                         |
| +- Output 2 -----+                     |
| | Type: [ WS2812B v ]                     |
| | Count: [ 60 ]                           |
| | Order: [ GRB v ]                         |
| +- Output 3 -----+                     |
| | Type: [ SK6812 v ]                       |
| | Count: [ 144 ]                           |
| | Order: [ RGBW v ]                       |
| +-----+                                 |
| [Save to child] [Clone to...]             |
+-----+
```

Page 4e — Edit Mode › Settings

Route	/edit/settings
Component	routes/tabs/Settings.svelte
Purpose	Network config, role switching, system actions.

Sections

- **Network** — SSID, password, hidden toggle, channel (Auto or 1–11)
- **Role** — Current role display + *Switch to Child* button (warning: requires factory rebind)
- **System** — Firmware version, Check for update, Restart, Factory reset

Layout

```
+-----+
| Settings                                     |
|                                             |
| NETWORK                                     |
| Name:      [ Loomx_Parent_4827      ]      |
| Password:  [ Loomx2847Blue          ]      |
| Hide SSID: [ OFF                    ]      |
| Channel:   [ Auto v ]                  |
|                                             |
| ROLE                                           |
| Current: Parent                             |
|           [ Switch to Child ]              |
|                                             |
| SYSTEM                                           |
| Firmware: v0.1.0                             |
| [ Check for update ] [ Restart ]            |
| [ Factory reset ]                             |
+-----+
```

Page 5 — Child Standby Screen

Route	/child
Component	routes/ChildStandby.svelte
Purpose	Minimal status display when controller is configured as Child.
Access	Connect directly to the child's IP (when in AP mode or via parent network).

Copy

- Title: **LumixLeaf**
- Status (connected): * *Connected to {parent_ssid}*
- Status (waiting): o *Waiting to be bound*
- Status (reconnecting): * *Reconnecting to {parent_ssid}...* (amber dot)
- Meta rows: **ID / Outputs / Signal**
- Action: **Switch to Parent role**

Layout

```
+-----+  
|                                     |  
|                               LoomX |  
|                                     |  
| LumixLeaf                         |  
|                                     |  
| * Connected to                     |  
|   Loomx_Parent_4827               |  
|                                     |  
| ID          AA:BB:CC:DD:EE:01    |  
| Outputs     4 configured            |  
| Signal      ||||                   |  
|                                     |  
  
| +-----+                          |  
| |       Switch to Parent role       | |  
| +-----+                          |  
+-----+
```

Reusable Components

Build these primitives first. All pages compose from this set.

Button

File	components/Button.svelte
Props	variant: 'primary' 'secondary' 'ghost' 'danger' · size: 'sm' 'md' 'lg' · disabled · icon · iconRight
Primary	bg #C6F542, text #000, hover bg #D8FA60
Secondary	bg transparent, border 0.5px #2A, text #FFF, hover bg #1A
Ghost	bg transparent, no border, text #888, hover text #FFF
Danger	bg transparent, border 0.5px #5A3030, text #F09595
Min height	44px (touch target)
Radius	14px

Card

File	components/Card.svelte
Props	selected, interactive, padding
Default	bg #0A, border 0.5px #2A, radius 16px, padding 20px
Selected	border 1px #C6F542
Interactive	cursor pointer, hover border #3A

Toggle

File	components/Toggle.svelte
Size	32x18px track, 14x14px thumb
Off state	track #2A, thumb white left
On state	track #C6F542, thumb #000 right, 200ms transition

Input

File	components/Input.svelte
Height	44px
Bg	#1A
Border	0.5px #2A, focus 1px #C6F542
Text	14px white, placeholder #555
Radius	12px
Padding	0 14px

StatusDot

File	components/StatusDot.svelte
Props	status: 'online' 'offline' 'pending' 'error'
Colors	online #5DCAA5 · offline #555 · pending #EF9F27 · error #F09595
Size	8px circle

SignalBars

File	components/SignalBars.svelte
Props	rssi (dBm) or strength (0-4)
Render	4 bars, filled #C6F542 when active, empty #2A
Thresholds	-50 = 4/4 · -65 = 3/4 · -75 = 2/4 · -85 = 1/4 · below = 0/4

Loader

3-dot pulse animation. Used on splash and during async actions. Each dot: 6px circle, accent green, opacity animates 30% -> 100% -> 30% with 200ms stagger across the three dots. Total loop 1.4s.

BrandMark

The LOOMX wordmark. Single component, accepts size prop ('sm' 16px, 'md' 22px, 'lg' 40px). Font: Helvetica-Bold or Inter-Black, letter-spacing -1px, color #C6F542.

Toast

Bottom-pinned notification. 14px white text on #1A surface, 0.5px border #2A, border-left 3px in semantic color (success #5DCAA5, error #F09595, info #C6F542). Auto-dismiss 2 seconds. Stacks vertically if multiple queued.

State Management & API

WebSocket Connection (Front-End -> Parent)

The UI maintains a single persistent WebSocket connection to the Parent at `ws://<host>/ws`. This connection is established on app mount and is the primary channel for state updates and commands.

Message Schema (Inbound from Parent)

```
// State snapshot (sent on connection)
{ "type": "snapshot", "children": [...], "presets": [...], "sequences": [...] }

// Children status update
{ "type": "children_update", "children": [
  { "id": "AA:BB:CC:01", "online": true, "rssi": -52, "name": "Child 1" }
]}

// Running sequence state
{ "type": "sequence_state", "sequence_id": "s_07", "step": 2, "running": true }

// Live state echo (when state changes)
{ "type": "state", "live": { "preset_id": "p_42", "running_sequence": null } }
```

Message Schema (Outbound from UI)

```
// Apply preset
{ "type": "preset_apply", "preset_id": "p_42" }

// Trigger sequence
{ "type": "sequence_trigger", "sequence_id": "s_07" }

// Stop all
{ "type": "stop_all" }

// Identify a child (flash white 3s)
{ "type": "identify", "child_id": "AA:BB:CC:01" }

// Live edit (during sequence builder preview)
{ "type": "live_set", "target": "all", "color": "#FF0000", "effect": "solid" }
```

REST API (Non-Realtime CRUD)

GET /api/role	Current role
POST /api/role	Set role · body: { role: 'parent' 'child' }
GET /api/network	Network config
POST /api/network	Update network · body: { ssid, password, hidden, channel }
POST /api/discover	Trigger child scan · returns array of LumixLeaf_* SSIDs
POST /api/bind	body: { ssids: [...] }

POST /api/unbind	body: { child_ids: [...] }
GET /api/children	List bound children
GET /api/children/:id /config	Get config
PUT /api/children/:id /config	Update config
POST /api/children/:id /push	Force config push
POST /api/children/:id /identify	Flash LEDs white 3s
POST /api/children/clone	body: { source_id, target_ids: [] }
GET /api/presets	List
POST /api/presets	Create (captures current live state)
PUT /api/presets/:id	Update
DELETE /api/presets/:id	Delete
GET /api/sequences	List
POST /api/sequences	Create
PUT /api/sequences/:id	Update
DELETE /api/sequences/:id	Delete
GET /api/sequences/:id/export	Download JSON file
POST /api/sequences/import	Upload JSON file
GET /api/system/info	Firmware version, uptime, free heap

POST /api/system/restart	Reboot
POST /api/system/factory-reset	Wipe NVS and reboot

Svelte Store Shape

```
// stores/children.js
import { writable } from 'svelte/store';
export const children = writable([
  // { id, name, online, rssi, outputs: [...], config: {...} }
]);

// stores/presets.js
export const presets = writable([
  // { id, name, states: { [child_id]: { color, effect, brightness } } }
]);

// stores/sequences.js
export const sequences = writable([
  // {
  //   id, name, loop,
  //   steps: [
  //     { id, target: 'all' | 'per_child', states: {...},
  //     effect, color, duration_ms, fade_ms }
  //   ]
  // }
]);

// stores/ws.js
export const wsState = writable({
  connected: false, reconnecting: false, last_error: null
});

// stores/live.js
export const live = writable({
  running_sequence: null, // sequence id if running
  current_step: 0,
  active_preset: null,
});
```

Interaction Patterns

Color Picker

- Inline expansion below the trigger — no separate page or full modal.
- 8 preset swatches + HSL slider + hex input.
- Preset swatches: white, red, orange, yellow, green, cyan, blue, magenta.
- Selecting a color: live-updates target immediately (during edit/builder) via WebSocket `live_set`.
- Closes on outside tap.

Destructive Confirmations

- Inline confirm — never browser `confirm()`.
- The destructive button transforms into two buttons: *Confirm delete* (red) + *Cancel*.
- Auto-reverts after 3 seconds if no action.

Toasts

- Triggered by: save success, config push complete, import success, error.
- Position: bottom center, 80px from bottom edge.
- Duration: 2 seconds (success) · 4 seconds (error).
- Stack: max 3 visible, new ones push old ones up.
- Tap to dismiss.

Pull to Refresh

- Available on Children tab — triggers a rescan.
- Loader appears at top during refresh.
- Min 800ms duration to feel deliberate.

Long Press Menus

- 500ms long press triggers contextual menu.
- Available on: sequence buttons, preset buttons, children list rows.
- Menu appears as a bottom sheet on mobile, popover on desktop.
- Haptic feedback on supported devices.

Live Indicators

- Sequence currently running: button shows pulsing accent border and small play icon overlay.

- Preset currently applied: button shows persistent accent border (no pulse).
- Child offline: row dot turns gray, name dims to #888.
- Child reconnecting: amber dot, gentle pulse.

Keyboard Shortcuts (Desktop Only)

Space	Stop all
Esc	Exit Edit Mode / close modal
E	Toggle Edit Mode
1–9	Trigger sequence N (View Mode)
■/Ctrl + S	Save (in Sequence Builder)

WLED UI Removal Checklist

The stock WLED UI must be completely removed. The following changes are required to the WLED firmware source tree.

Files to Delete

```
wled00/data/index.htm
wled00/data/index.js
wled00/data/index.css
wled00/data/iro.js
wled00/data/rangetouch.js
wled00/data/settings.htm
wled00/data/settings_*.htm      (all settings_*.htm variants)
wled00/data/pixart/            (entire directory)
wled00/data/welcome.htm
wled00/data/liveview.htm
wled00/data/liveviewws.htm
wled00/data/cpal/              (entire color palette UI directory)
wled00/data/style.css
wled00/data/common.js
wled00/html_*.h                (regenerated headers – delete the originals)
```

Files to Keep but Modify

```
wled00/wled_server.cpp          // Replace all UI routes with single bundle route
wled00/wled_server.h
wled00/ws.cpp                   // Extend WebSocket handler with Loomx message schema
wled00/cfg.cpp                  // Add NVS keys for role, parent_creds, child_bindings
wled00/improv.cpp               // Keep for serial provisioning (optional)
platformio.ini                  // Add web-ui build step + bundle-to-header script
```

New Files to Add

```
wled00/loomx/role.cpp           // Parent/Child role logic
wled00/loomx/role.h
wled00/loomx/ws_protocol.cpp     // Loomx WebSocket message schema
wled00/loomx/ws_protocol.h
wled00/loomx/discovery.cpp       // Child discovery + binding
wled00/loomx/discovery.h
wled00/loomx/storage.cpp         // LittleFS for presets/sequences/configs
wled00/loomx/storage.h
wled00/web_ui_bundle.h           // Generated: gzipped UI bundle as byte array
```

HTTP Route Replacement

Replace the entire WLED HTTP routing setup with a minimal one: serve the bundled UI at /, expose /api/* for REST, and /ws for the WebSocket. All other WLED routes (/settings, /json, /liveview, etc.) are removed.

```
// wled_server.cpp – REPLACE entirely
void initServer() {
    // Single UI route
    server.on("/", HTTP_GET, [](AsyncWebServerRequest *r) {
        auto *resp = r->beginResponse_P(200, "text/html",
```

```

                                web_ui_bundle, web_ui_bundle_len);
    resp->addHeader("Content-Encoding", "gzip");
    r->send(resp);
});

// REST API
loomx_register_api(&server);

// WebSocket
ws.onEvent(loomx_ws_event);
server.addHandler(&ws);

// 404 fallback (SPA – return index for any unmatched GET)
server.onNotFound([](AsyncWebServerRequest *r) {
    if (r->method() == HTTP_GET) {
        auto *resp = r->beginResponse_P(200, "text/html",
                                web_ui_bundle, web_ui_bundle_len);
        resp->addHeader("Content-Encoding", "gzip");
        r->send(resp);
    } else {
        r->send(404);
    }
});
}

```


Build Order

Recommended implementation sequence. Each milestone should be runnable and demoable before moving to the next.

M1 — Project Scaffold

Set up Vite + Svelte project in web-ui/. Configure CSS tokens. Build BrandMark + Loader components. Get the splash screen rendering and animating.

M2 — Firmware Bundle Pipeline

Wire up the Vite build to output a gzipped bundle. Write the bundle-to-header script. Modify the firmware HTTP route to serve the bundle. Verify the splash screen loads from the ESP32.

M3 — WLED UI Removal

Delete all files listed in the WLED Removal Checklist. Strip routes. Confirm the device boots and serves only the Loomx UI.

M4 — Role Selection + Setup Wizard

Build Page 1 and Page 2. Persist role and network config to NVS. Implement credential auto-generation logic.

M5 — WebSocket Infrastructure

Establish the persistent WebSocket connection from UI to Parent. Implement message routing, reconnection logic, snapshot delivery on connect.

M6 — Children Tab + Discovery + Binding

Page 4c. Implement scan, discovery, multi-select bind. Build StatusDot, SignalBars components. Live updates of online/offline state.

M7 — Configs Tab

Page 4d. Per-output LED type/count/order. Clone flow. Export/import JSON. Verify GPIO numbers never appear in any UI surface.

M8 — Presets

Page 4b. Capture, apply, rename, delete. WebSocket apply broadcasts to all children.

M9 — Sequence Engine + Builder

Page 4a and 4a.1. Visual builder, step ordering, drag handle, live preview during edit. Lockstep execution across children via `exec_at` timestamps.

M10 — View Mode

Page 3. Hotkey grid. Long-press menus. Stop All. Live running-sequence indicator.

M11 — Settings

Page 4e. Network editor, role switcher, factory reset, restart.

M12 — Child Standby

Page 5. Status display, role-switch UI when accessed directly.

M13 — Polish Pass

Toasts, error states, empty states, reduced motion handling, keyboard shortcuts, accessibility audit (focus rings, ARIA labels, screen reader).

M14 — Acceptance Test

Run the Acceptance Criteria checklist (next section) end-to-end on real hardware.

Acceptance Criteria

The build is complete when all the following are true on real hardware.

Functional

- Fresh-flashed device boots to splash, then role selection.
- Selecting Parent leads to setup wizard with auto-generated credentials in the documented format.
- Parent boots into AP mode broadcasting the configured SSID.
- Children boot into AP mode broadcasting LumixLeaf_NNNN.
- Parent's Scan finds all nearby LumixLeaf APs and lists them.
- Multi-select bind successfully transfers credentials to all selected children; they join the Parent's network.
- Smart Config Handling: re-binding a previously-configured child does NOT overwrite its config.
- Configuring outputs: GPIO numbers never appear anywhere in the UI.
- Saving a preset captures all bound children's current state; applying it restores them simultaneously.
- Triggering a sequence executes all steps in lockstep across all children — no perceptible drift.
- Sequence export downloads a valid JSON file named after the sequence.
- Importing that same JSON on another Parent reproduces the sequence exactly.
- Stop All immediately turns off all children.
- Identify flashes a single specified child white for 3 seconds.

Visual / Brand

- Every page background is pure black (#000) — no off-black, no gradients.
- Neon green (#C6F542) appears only on: brand mark, primary buttons, active states, success indicators, focus rings.
- All touch targets ≥ 44 px tall.
- Splash screen shows for minimum 1500ms.
- Loader 3-dot animation runs as specified.
- Active sequence button pulses; applied preset shows solid accent border.
- Reduced motion preference disables all loops; transitions remain.

Performance

- Total UI bundle ≤ 80 KB gzipped.
- First paint within 200ms of HTTP response on iPad Safari on the Parent's network.
- WebSocket message round-trip ≤ 100 ms on Parent's AP.
- Sequence sync drift across 4 children ≤ 30 ms when measured with high-speed camera.

Robustness

- WebSocket auto-reconnects with exponential backoff (capped at 5s) on disconnect.
- Refreshing the browser does not lose state — re-fetches snapshot on connect.
- Killing and restoring power to a child: it reconnects, retains its config, resumes responding.
- Killing power to the Parent: children retain their last-known config and idle off until Parent returns.
- Factory reset wipes NVS and LittleFS, returns device to first-boot state.

Out of Scope (Confirm with Stakeholder Before Adding)

- Cloud sync or remote control over internet.
- Native mobile apps.
- Multi-Parent meshing.
- 5 GHz WiFi support (hardware limitation).
- User accounts or authentication beyond the WiFi password.

Appendix

Example Sequence JSON

```
{
  "schema_version": 1,
  "id": "s_countdown_001",
  "name": "Countdown",
  "loop": false,
  "created_at": "2026-05-21T18:00:00Z",
  "steps": [
    {
      "id": "step_1",
      "target": "all",
      "effect": "solid",
      "color": "#FFFFFF",
      "brightness": 200,
      "duration_ms": 3000,
      "fade_ms": 200
    },
    {
      "id": "step_2",
      "target": "all",
      "effect": "flash",
      "color": "#FF0000",
      "brightness": 255,
      "duration_ms": 2000,
      "fade_ms": 0,
      "effect_params": { "flash_rate_hz": 4 }
    },
    {
      "id": "step_3",
      "target": "per_child",
      "states": {
        "AA:BB:CC:01": { "color": "#0000FF", "effect": "solid" },
        "AA:BB:CC:02": { "color": "#00FF00", "effect": "solid" },
        "AA:BB:CC:03": { "color": "#FFFFFF", "effect": "solid" },
        "AA:BB:CC:04": { "color": "#FF00FF", "effect": "solid" }
      },
      "duration_ms": 5000,
      "fade_ms": 1000
    }
  ]
}
```

Example Config JSON

```
{
  "schema_version": 1,
  "child_id": "AA:BB:CC:01",
  "name": "Child 1",
  "outputs": [
    { "slot": 1, "led_type": "WS2812B", "count": 60, "color_order": "GRB" },
    { "slot": 2, "led_type": "WS2812B", "count": 60, "color_order": "GRB" },
    { "slot": 3, "led_type": "SK6812", "count": 144, "color_order": "RGBW" },
    { "slot": 4, "led_type": "WS2811", "count": 50, "color_order": "BRG" }
  ]
}
```

```
}
```

CSS Tokens (Drop-In)

```
:root {
  /* Backgrounds */
  --bg-base: #000000;
  --bg-surface: #0A0A0A;
  --bg-surface-2: #1A1A1A;

  /* Borders */
  --border-default: #2A2A2A;
  --border-strong: #3A3A3A;
  --border-active: #C6F542;

  /* Text */
  --text-primary: #FFFFFF;
  --text-secondary: #888888;
  --text-tertiary: #444444;

  /* Brand */
  --accent: #C6F542;
  --accent-ink: #000000;
  --accent-hover: #D8FA60;

  /* Semantic */
  --success: #5DCAA5;
  --warning: #EF9F27;
  --danger: #F09595;

  /* Radii */
  --radius-input: 12px;
  --radius-button: 14px;
  --radius-card: 16px;
  --radius-pill: 999px;

  /* Type */
  --font-sans: 'Inter', system-ui, -apple-system, sans-serif;
  --font-mono: 'JetBrains Mono', ui-monospace, monospace;

  /* Spacing */
  --sp-1: 4px;
  --sp-2: 8px;
  --sp-3: 12px;
  --sp-4: 16px;
  --sp-5: 20px;
  --sp-6: 24px;
  --sp-7: 32px;
  --sp-8: 48px;

  /* Motion */
  --t-fast: 150ms;
  --t-base: 200ms;
  --ease-out: cubic-bezier(0.16, 1, 0.3, 1);
}

* { box-sizing: border-box; }

html, body {
  margin: 0;
```

```

background: var(--bg-base);
color: var(--text-primary);
font-family: var(--font-sans);
font-size: 14px;
line-height: 1.5;
-webkit-font-smoothing: antialiased;
overscroll-behavior: none;
}

button {
  font-family: inherit;
  cursor: pointer;
  min-height: 44px;
  border-radius: var(--radius-button);
  border: 0.5px solid var(--border-default);
  background: transparent;
  color: var(--text-primary);
  transition: all var(--t-fast) var(--ease-out);
}

button.primary {
  background: var(--accent);
  color: var(--accent-ink);
  border-color: var(--accent);
  font-weight: 500;
}

button.primary:hover { background: var(--accent-hover); }

input, select, textarea {
  background: var(--bg-surface-2);
  border: 0.5px solid var(--border-default);
  border-radius: var(--radius-input);
  color: var(--text-primary);
  padding: 0 14px;
  height: 44px;
  font-family: inherit;
  font-size: 14px;
}

input:focus, select:focus, textarea:focus {
  outline: none;
  border-color: var(--accent);
  box-shadow: 0 0 0 1px var(--accent);
}

@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    transition-duration: 0.01ms !important;
  }
}

```

Companion Documents

- **loomx_claude_code_prompt.md** — Full firmware + system architecture spec
- **loomx_ui_flow.md** — UI flow + ASCII page mockups (source for this PDF)